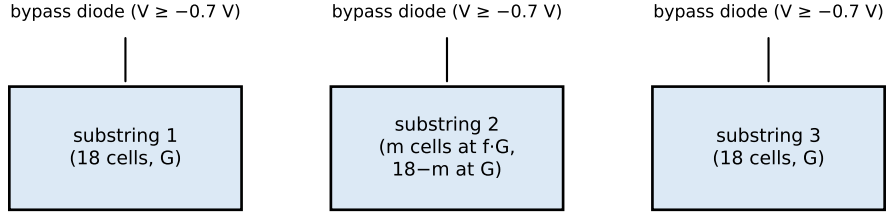

$$V_{string}(I) = \Sigma_k \max(-0.7\ V, \Sigma_{groups} V_{group}(I)), \quad V_{group}(I) = R_{sh}(I_{ph} + I_0) - I(R_s + R_{sh}) - a \cdot W(I_0 R_{sh} / a \cdot \exp(R_{sh}(I_{ph} + I_0 - I)/$$